

## **Project Name: AI Code Navigator (Backend) (week – 8 and 9)**

### **Problem: Fixing Duplicate Data in AI Code Navigator**

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#### **What Was the Problem?**

When I uploaded the same file or the same GitHub repository more than once, the system was saving it again and again in the database. This was a problem because:

- The database was getting filled with the same data multiple times
  - It was calling the OpenAI API every single time even for files it already had which wastes money
  - Search results could show the same file multiple times
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#### **How Did We Fix It?**

The idea was simple before saving any file, first check if we already have it. To do that, we used **hashing**.

A hash is like a unique fingerprint for a file. If the content of two files is exactly the same, their hash will also be exactly the same. If even one character is different, the hash will be completely different.

We used Node.js's built-in crypto module (no new packages needed) to generate a SHA256 hash of the code content.

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#### **What Files We Changed**

##### **1. models/code.js**

We added a new field called `contentHash` to the database schema. This stores the fingerprint of each file. We also made it unique so the database itself will never allow two documents with the same hash.

##### **2. controllers/codecontroller.js**

We added a small helper function at the top:

```
const hashCode = (code) => crypto.createHash('sha256').update(code).digest('hex');
```

Then in all 3 upload functions (processCode, uploadCode, uploadRepo), we added the same logic:

1. Compute the hash of the code
2. Check if that hash already exists in MongoDB
3. If yes skip it, return the existing summary (no OpenAI call)
4. If no process it normally, save it with the hash

The important thing is we check the hash **before** calling OpenAI This means no wasted API calls for duplicates.

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## How We Tested It

We tested all 4 API endpoints:

- Sent the same code twice to /api/process-code 2nd time returned duplicate: true
- Uploaded the same file twice to /api/upload 2nd time returned duplicate: true
- Uploaded the same GitHub repo twice to /api/upload-repo 1st time saved 8 files, 2nd time skipped all 8
- Tested /api/search → returned the correct file with a correct answer

All tests passed The database now only stores unique files

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## Result

Before the fix, uploading the same repo 3 times would create 3 copies of every file in the database and make 3x the OpenAI API calls.

After the fix, the first upload works normally. Every upload after that is detected and skipped instantly.